

Yüksel Kaan Bölükbaş

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Education

Hacettepe University – B.Sc. in Artificial Intelligence Engineering, 4rd year Ankara, Turkey Expected 2027

Experience

Datateam Information Technologies – Intern Ankara, Turkey July 2025 – August 2025

- Developed an AI-powered global supplier platform by processing **3 million+ filtered records** spanning **25 years** and **42 countries**.
- Implemented automated classification and big data analysis using **PostgreSQL** and **GPT-4o mini**, optimizing supplier audit workflows.

Nart Information Technologies – Intern Ankara, Turkey February 2026 – March 2026

- Built an **hourly trip count forecasting** pipeline for **NYC Yellow Taxi** dataset using **XGBoost, LightGBM, Prophet, and Ridge Regression**, with best model achieving **$R^2=0.979$** (XGBoost); engineered lag features (up to lag-336), rolling averages, cyclical encodings, and weather data via Open-Meteo API.
- Processed large-scale **Parquet** files using **DuckDB** and optimized all models through **Optuna** hyperparameter tuning with time-series cross-validation.

Hacettepe University UAV Society – Director, Project Coordination Unit Ankara, Turkey August 2025 – February 2026

- Supervised multiple **TEKNOFEST teams** and coordinated member selection processes, ensuring high-performance standards across the society.

Hacettepe University UAV Society – Vice President Ankara, Turkey February 2026 – Present

- Leading strategic planning and operational coordination of the society, overseeing **TEKNOFEST** project teams and representing the community in university-level activities.

Projects

Simurgh Fly – TEKNOFEST International UAV Competition Ankara, Turkey
Team Lead (Captain) February 2026 – Present

- Leading a multidisciplinary team of **10 members** (Software, Electronics, and Mechanics) to develop an autonomous fixed-wing UAV for AI-driven disaster response and aid distribution.
- Implementing real-time object detection using **YOLOv11** on **Raspberry Pi 5 with AI Kit**, achieving high-precision human detection and density analysis via **DBSCAN** clustering algorithms.

Simurgh Fly – TEKNOFEST Fighting UAV (Savaşan İHA) – Software Team Lead Ankara, Turkey August 2025 – February 2026

- Led a team of **10 software engineers** to develop autonomous systems for a combat UAV, focusing on **Computer Vision, ROS, and PX4**.
- Achieved **mAP50 of 0.96** and **mAP50-95 of 0.80**; managed end-to-end integration and real-time testing in **Gazebo** ensuring **95% recall**.

Simurgh Fly – TEKNOFEST UAV Project – Computer Vision Specialist Ankara, Turkey November 2024 – August 2025

- Trained deep learning models on **60,000+ aerial images**, achieving **94% accuracy**; optimized inference on **NVIDIA Jetson Xavier NX** via **CUDA** and **TensorRT** to **45+ FPS**.

Operator Audio Monitoring ML Model – Machine Learning Project Remote March 2024 – November 2024

- Built an end-to-end **ML pipeline** for real-time operator monitoring via **Audio Signal Processing**, including **profanity detection, WPM tracking, and aggression detection** through frequency spectrum analysis.

TEKNOFEST Robotaxi – Passenger Autonomous Vehicle – Software Member Ankara, Turkey April 2025 – October 2025

- Conducted target detection model testing and sensor fusion simulations in the **Gazebo** environment.

Technical Skills

Languages & Frameworks: Python, Java, C++, Machine Learning, PyTorch, YOLO, Computer Vision, PostgreSQL, ROS, Streamlit, Linux, CUDA, TensorRT, Git.